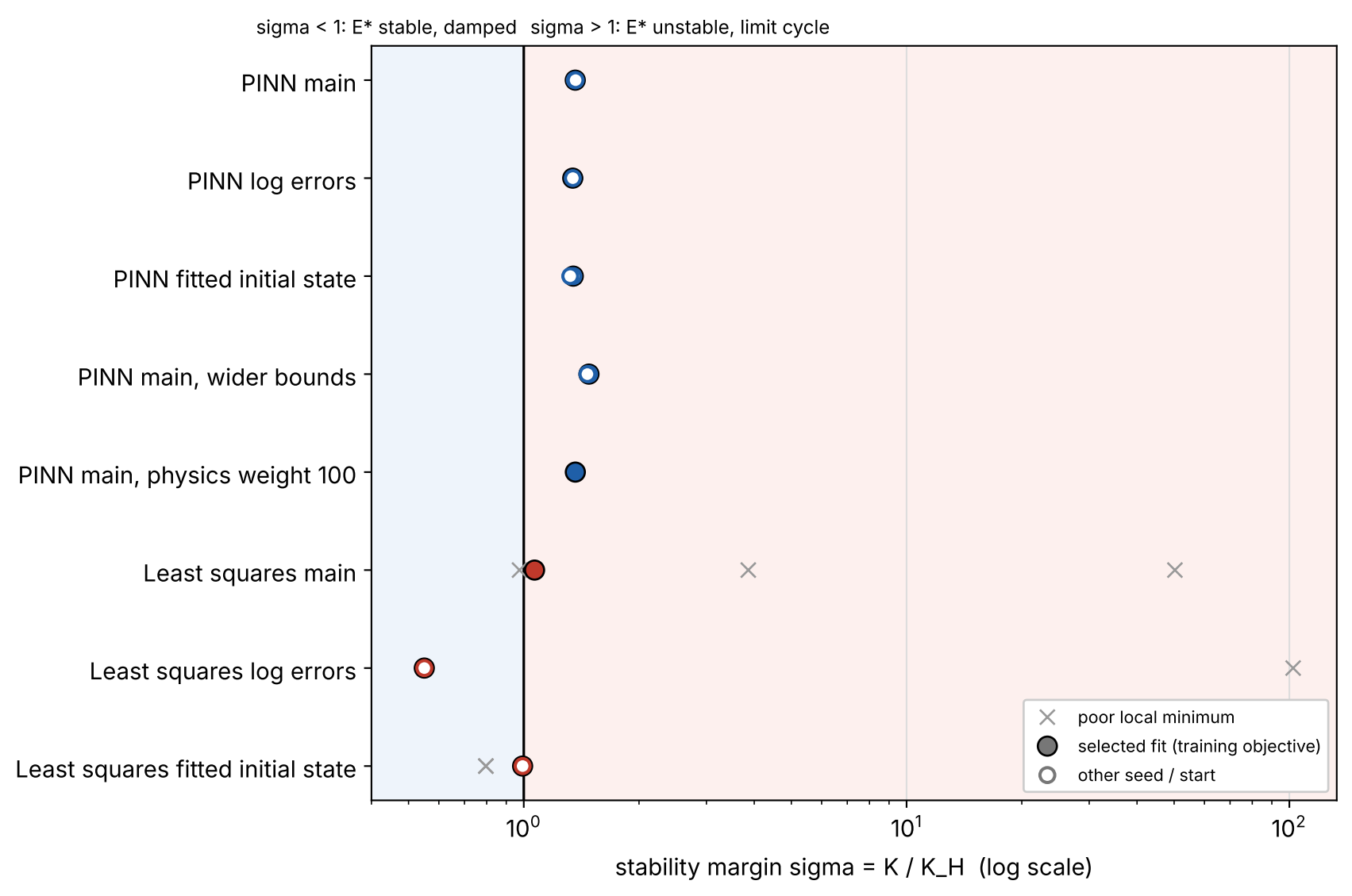
sigma < 1: E* stable, damped
sigma > 1: E* unstable, limit cycle
PINN main
PINN log errors
PINN fitted initial state
PINN main, wider bounds
PINN main, physics weight 100
Least squares main
Least squares log errors
Least squares fitted initial state
poor local minimum
selected fit (training objective)
other seed / start
$10^0$
$10^1$
$10^2$
stability margin sigma = K / K_H (log scale)